25th National High School Physics Competition: Final Solutions

1.

1. Method 1: intercept at C on Ax, time t.

Let l be the $A-C$ distance and l_p the distance from B to C (Fig. 1). Then

$$l = vt \ , \ l_p = v_p t \qquad (1)$$

and

$$l_p^2 = d^2 + l^2 - 2dl\cos\theta . \qquad (2)$$

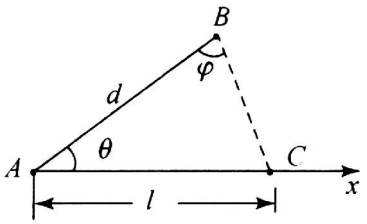


Fig. 1

Solving (1), (2) gives

$$l = \frac{d}{1-(v_p/v)^2}\{\cos\theta \pm [(\frac{v_p}{v})^2 - \sin^2\theta]^{1/2}\}. \qquad (3)$$

In (3), interception requires real roots, giving the necessary condition

$$v_p \geqslant v\sin\theta \ . \qquad (4)$$

Method 2: let φ be the angle between BA and BC (Fig. 1). The sine rule gives

$$\frac{l_p}{\sin\theta} = \frac{l}{\sin\varphi} \ .$$

Using (1) gives

$$\frac{v_p}{v} = \frac{\sin\theta}{\sin\varphi} \ .$$

$\sin\varphi \leqslant 1$ gives the necessary condition (4).

2. Let l_{min} be the least interception distance from A. From (3),

$$l_{min} = \frac{d}{1-(v_p/v)^2}\{\cos\theta \pm [(\frac{v_p}{v})^2 - \sin^2\theta]^{1/2}\}. \qquad (5)$$

If the receiver is closer to A than l_{min}, he gets the ball first. The condition for avoiding interception is

$$l_r < l_{min} \ . \qquad (6)$$

From (5), (6),

$$l_r < \frac{d}{1-(v_p/v)^2}\{\cos\theta \pm [(\frac{v_p}{v})^2 - \sin^2\theta]^{1/2}\} \qquad (7)$$

By (7), if a receiver on Ax, distance l_r from A, satisfies this inequality, the defender initially at B cannot intercept the ball.

3. Method 1: before the defender at B reaches C_1 on Ax, distance l_{min} from A,

if the receiver gets closer to A than $l_{\min}$, the defender at B cannot intercept (Fig. 2). If $L \leqslant l_{\min}$, it is part 2. If $L > l_{\min}$, let the receiver be at E on Ax.

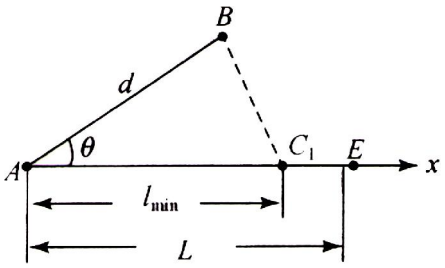


Fig. 2

His time to reach C_1 is

$$t_{rm} = (L - l_{\min}) / v_r \ ; \qquad (8)$$

Defender arrival time at C_1: $t_{pm} = (d^2 + l_{\min}^2 - 2dl_{\min}\cos\theta)^{1/2} / v_p$.

The condition for avoiding interception is

$$t_{rm} < t_{pm} \ . \qquad (9)$$

Thus $$L < \frac{v_r}{v_p}(d^2 + l_{\min}^2 - 2dl_{\min}\cos\theta)^{1/2} + l_{\min} , \qquad (10)$$

where $l_{\min}$ is given by (5).

Method 2: the defender and ball reach C_1 at the same time, so

$$t_{pm} = l_{\min} / v \ .$$

The no-interception condition (9) and equation (8) give

$$L < (1 + v_r / v) l_{\min} \qquad (11)$$

Again $l_{\min}$ is given by (5). Equations (11) and (10) clearly agree.

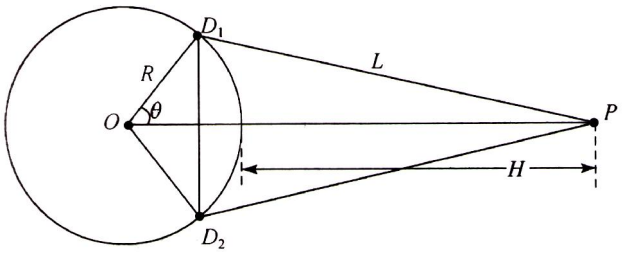
2.

1. (I) In a chosen frame, satellite motion gives its coordinates x, y, z at every instant. A signal emitted at t reaches ground station i at t_i. The signal travels at light speed c, hence

$$(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = c^2 (t - t_i)^2 \qquad (i = 1, 2, 3), \qquad (1)$$

x_i, y_i, z_i are known, premeasured station i coordinates; its clock measures t_i. The signal supplies t, also known. Equation (1) thus has three unknowns, x, y, z, requiring three independent equations: at least three ground stations. These are (1) for $i = 1, 2, 3$.

(II) (i) Earth centre O and stations D_1, D_2 form an isosceles triangle; the equal sides are R. Thus



At emission, the satellite distances to both stations are equal:

$$L = c\tau \ . \qquad (2)$$

When satellite P is in the triangle plane, its altitude is maximal, H. Let θ be the latitude of D_1 , D_2. The cosine rule gives

$$L^2 = R^2 + (H+R)^2 - 2R(H+R)\cos\theta \ . \qquad (3)$$

Equations (2), (3) give

$$H = \sqrt{(c\tau)^2 - (R\sin\theta)^2} - R(1-\cos\theta) \ . \qquad (4)$$

Equation (4) also follows directly from the diagram.

(ii) A small latitude error $\triangle\theta$ gives, by (3), an error $\triangle H$ in H. Then we have

$$L^2 = R^2 + (H+\triangle H+R)^2 - 2R(H+\triangle H+R)\cos(\theta+\triangle\theta) \ . \qquad (5)$$

Expand (5). Small θ gives small $\triangle H$. Drop higher orders and subtract (3), giving

$$\triangle H = -\frac{R(R+H)\sin\theta\triangle\theta}{H+(1-\cos\theta)R} \ , \qquad (6)$$

Here H is given by (4).

(iii) An error $\triangle\tau$ in time τ gives an error in L:

$$\triangle L = c\triangle\tau \ . \qquad (7)$$

By (3), this gives an error $\triangle H$ in H. Then

$$(L+\triangle L)^2 = R^2 + (H+\triangle H+R)^2 - 2R(H+\triangle H+R)\cos\theta. \qquad (8)$$

From (7), (8) and 3), neglect higher orders to obtain

$$\triangle H = \frac{c^2\tau\triangle\tau}{H+R(1-\cos\theta)} \ , \qquad (9)$$

Here H is given by (4).

2. (i) From 4), $H = 2.8\times10^{4}$ km. (ii) From 6), $\triangle H = \mp 25$m. (iii) Substituting into 9) gives $H = \pm 3$m .

3. In a chosen frame, let position be x , y , z and time t. Satellite i has coordinates x_i , y_i , z_i at time t_i. Its signal travels at light speed, so

$$(x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2 = c^2(t-t_i)^2 \qquad (i = 1 \ , \ 2 \ , \ 3 \ , \ 4), \qquad (10)$$

(1) has four unknowns t , x , y , z, requiring four independent equations. Receive four or more satellites at once. Position and time follow from (10), i

= 1 , 2 , 3 , 4: four independent equations.

4. (I) The satellite clock slows by $[1-(v/c)^2]^{1/2}$. The ground reading T minus satellite reading t is

$$T-t=T-\sqrt{1-(\frac{v}{c})^2}\,T=[\,1-\sqrt{1-(\frac{v}{c})^2}\,]\,T\ , \qquad (11)$$

Here v is satellite speed relative to the ground, determined by

$$\frac{v^2}{r}=\frac{GM}{r^2}\ , \qquad (12)$$

G is the gravitational constant, M Earth's mass, r orbit radius. From (11),

$$v=\sqrt{\frac{GM}{r}}=\sqrt{\frac{g}{r}}R=\sqrt{\frac{g}{R+h}}R\ ,$$

Here R is Earth's radius, h altitude and $g=GM/R^2$ ground gravity. Substitution gives $v=$ 3489 , hence $(v/c)^2\approx$ 1.68 $^{-10}$, which is very small. Thus

$$[\,1-(\frac{v}{c})^2\,]^{1/2}\approx 1-\frac{1}{2}(\frac{v}{c})^2\ .$$

Finally the clock difference over 24 is

$$T-t\approx\frac{1}{2}(\frac{v}{c})^2T=\frac{1}{2}\,\frac{gR^2}{c^2(R+h)}T=7.3\,\mu\text{s}\ . \qquad (13)$$

(II) The satellite reading t minus the inertial-clock reading T_0 at infinity is

$$t-T_0=\sqrt{1-2\frac{\phi}{c^2}}T_0-T_0=(\sqrt{1-2\frac{\phi}{c^2}}-1)T_0\ . \qquad (14)$$

The gravitational potential magnitude at the satellite clock is

$$\phi=\frac{GM}{R+h}=\frac{R^2}{R+h}g\ . \qquad (15)$$

Thus $\dfrac{\phi}{c^2}=\dfrac{gR^2}{c^2(R+h)}$;

Numerically, $\phi/c^2=$ 1.68 $^{-10}$ is very small, so equation (14) becomes

$$t-T_0\approx-\frac{\phi}{c^2}T_0\ . \qquad (16)$$

Similarly, ground reading T minus the inertial clock at infinity is

$$T-T_0=\sqrt{1-2\frac{\phi_E}{c^2}}T_0-T_0=(\sqrt{1-2\frac{\phi_E}{c^2}}-1)T_0\ . \qquad (17)$$

The gravitational potential magnitude at the ground clock is

$$\phi_E=\frac{GM}{R}=gR\ . \qquad (18)$$

Thus $\frac{\phi_E}{c^2} = \frac{gR}{c^2}$;

Numerically, $\phi_E / c^2 = 6.96 \times 10^{-10}$ is very small. As above, (17) is approximately given by

$$T - T_0 \approx -\frac{\phi_E}{c^2} T_0 \ . \qquad (19)$$

Subtract (16), (19) to find satellite minus ground clock readings:

$$t - T \approx -\frac{\phi - \phi_E}{c^2} T_0 \ . \qquad (20)$$

Solve (19) for T_0 and substitute into (20):

$$t - T \approx -\frac{\phi - \phi_E}{c^2} / (1 - \frac{\phi_E}{c^2}) T \approx -\frac{\phi - \phi_E}{c^2} T = \frac{gR}{c^2} \frac{h}{R+h} T. \qquad (21)$$

The stated 24 is ground elapsed time T. Finally, after 24 , the satellite minus ground reading is

$$t - T = 46 \mu s \ . \qquad (22)$$

3.

1. For constant room temperature, the heat pump output must equal the room heat loss rate. Let q be the indoor heat output and P the electric input. The heat extracted outdoors at low temperature per unit time is $q-P$. Then

$$\frac{q-P}{P} \leqslant \frac{T_2}{T_1 - T_2} \ , \qquad (1)$$

T_1 is the indoor hot absolute temperature, T_2 the outdoor one. From (1),

$$P \geqslant \frac{T_1 - T_2}{T_1} q \ . \qquad (2)$$

Minimum cost requires minimum P: equality in " " in (2), the ideal case, minimizes P. Thus the least electric power is

$$P_{\min} = \frac{T_1 - T_2}{T_1} q \ . \qquad (3)$$

Heat conducts out of the room through the glass. The loss rate is

$$H = k \frac{T_1 - T_2}{l} S \ . \qquad (4)$$

For a constant room temperature,

$$q = H \ . \qquad (5)$$

Equations (3) ~ (5) give

$$P_{\min} = k\frac{S(T_1 - T_2)^2}{lT_1} \quad . \qquad (6)$$

For operating time t and price per kWh c, the least electricity cost is

$$C_{\min} = P_{\min} tc \quad . \qquad (7)$$

Note: $T_1 = 20.00\ \mathrm{K} + 273.15\ \mathrm{K} = 293.15\ \mathrm{K}$, $T_2 = -5.00\ \mathrm{K} + 273.15\ \mathrm{K} = 268.15\ \mathrm{K}$, 1 unit=1kW . Equations (6), (7), with the stated data, give

$$C_{\min} = \frac{(T_1 - T_2)^2}{T_1 l} Sktc = 23.99 \text{ yuan.} \qquad (8)$$

Thus, ideally, this heat pump costs about 24 yuan for 12 .

2. Let T_i, T_0 be the inner and outer air-layer surface temperatures. The rates of heat conduction through the inner glass, air and outer glass are respectively

$$H_1 = k\frac{T_1 - T_i}{l}S \quad , \qquad (9)$$

$$H_2 = k_0\frac{T_i - T_0}{l_0}S \quad , \qquad (10)$$

$$H_3 = k\frac{T_0 - T_2}{l}S \quad . \qquad (11)$$

In steady heat conduction,

$$H_1 = H_2 = H_3 \quad . \qquad (12)$$

Equations (9) ～ (12) give

$$k\frac{T_1 - T_i}{l} = k_0\frac{T_i - T_0}{l_0} \quad \text{and} \quad T_1 - T_i = T_0 - T_2 \quad . \qquad (13)$$

Solving (13) gives

$$T_i = \frac{l_0 k + lk_0}{l_0 k + 2lk_0}T_1 + \frac{lk_0}{l_0 k + 2lk_0}T_2 \quad . \qquad (14)$$

Substitute (14) into 9):

$$H_1 = \frac{kk_0}{l_0 k + 2lk_0}(T_1 - T_2)S \quad . \qquad (15)$$

Constant temperature requires $q = H_1$. By (3), for double glazing the minimum electric power is

$$P'_{\min} = \frac{kk_0}{l_0 k + 2lk_0}\frac{(T_1 - T_2)^2}{T_1}S \quad . \qquad (16)$$

Ideally, the cost for operation time t is

$$C'_{\min} = P'_{\min} tc \quad ; \qquad (17)$$

Substituting the data gives

$$C'_{\min} = 2.52 \text{ yuan.} \qquad (18)$$

Thus the selected double glazing saves the following cost over 12 :

$$\triangle C_{\min} = C_{\min} - C'_{\min} = 21.47 \text{ yuan.} \qquad (19)$$

4.

1. First let one electron tunnel from plate A to B. Let Q be the charge on plate A before tunnelling, and V_{AB} the plate voltage (problem Fig. 3). Then

$$V_{AB} = Q / C \ . \qquad (1)$$

The stored capacitor energy is

$$U = \frac{1}{2} C V^2_{AB} \ . \qquad (2)$$

After the electron tunnels to B, the charge on A is

$$Q' = Q + e \ , \qquad (3)$$

Here e is electron charge magnitude. The new voltage and stored energy are

$$V'_{AB} = \frac{Q + e}{C} \ , \ U' = \frac{1}{2} C V'^2_{AB} \ . \qquad (4)$$

Coulomb blockade, forbidding tunnelling, requires

$$U' - U > 0 \ . \qquad (5)$$

Equations (1) ~ (5) give

$$V_{AB} > -\frac{1}{2}\frac{e}{C} \ . \qquad (6)$$

Now let an electron tunnel from B to A. Again Q is the charge on A before tunnelling, V_{AB} the plate voltage. When it tunnels from B to A, plate A has charge $Q' = Q - e$. Similarly, forbidding tunnelling from B to A requires the following condition:

$$V_{AB} < \frac{1}{2}\frac{e}{C} \ . \qquad (7)$$

From (6), (7), voltage V_{AB} between $-e/2C \sim e/2C$ blocks single-electron tunnelling. The blockade condition is

$$-\frac{1}{2}\frac{e}{C} < V_{AB} < \frac{1}{2}\frac{e}{C} \ . \qquad (8)$$

2. By the stated condition and (8), the tunnelling threshold satisfies

$$V_{AB} = \frac{1}{2}\frac{e}{C} = 0.10 \text{ mV} \ . \qquad (9)$$

Substituting the data into (9) gives

$$C = 8.0 \times 10^{-16} \text{ F} \ . \qquad (10)$$

3. In Fig. 3, the left MIM capacitance is C_S; the right MIM has capacitance C_D. In Fig. a, Q_1 , Q_2 are charges on C_S, C_D. Thus the net island charge is given by

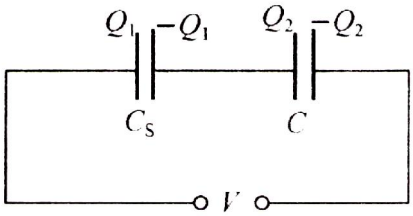


Fig. a

$$-ne = Q_2 - Q_1 . \qquad (11)$$

The electrostatic energy is the sum of that in C_S and C_D:

$$U = \frac{Q_1^2}{2C_S} + \frac{Q_2^2}{2C_D} ; \qquad (12)$$

The voltage is

$$V = \frac{Q_1}{C_S} + \frac{Q_2}{C_D} . \qquad (13)$$

Solving (11) ~ (13) gives

$$U = \frac{1}{2}CV^2 + \frac{(Q_2 - Q_1)^2}{2(C_S + C_D)} . \qquad (14)$$

With constant V, (13) gives the energy due only to net island charge: $U_n = (-ne)^2 / 2(C_S + C_D)$.

4. The plot of U_n against $C_G V_G$ is Fig. b; Table 2 gives the $C_G V_G / e$ ranges.

Table 2

n	0	1	2	3
$C_G V_G / e$ range	0～0.5	0.5～1.5	1.5～2.5	2.5～3.0

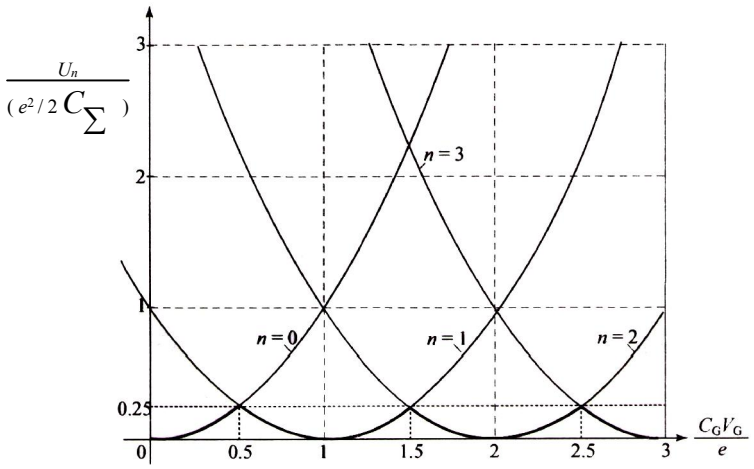


Fig. b

5.

1. In Fig. 1, z is normal to AB. Two parallel rays enter plane AB at C, C', with incidence angle i. The refraction angle is r. The two refracted rays reach the surface at D, D', at incidence angles i_2, i'_2. In $\triangle OCD$, the distance from O to entry C is y_c. By the sine rule,

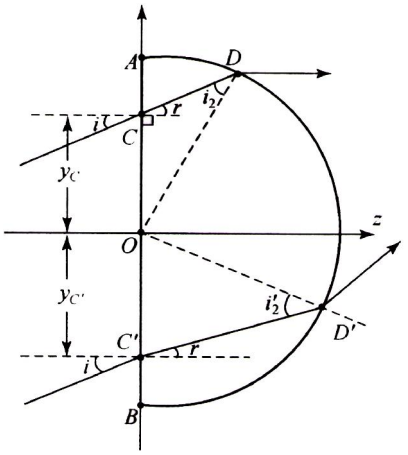


Fig. 1

$$\frac{y_c}{\sin i_2} = \frac{R}{\sin(90^\circ + r)}\ ,\ \text{so}\quad y_c = \frac{\sin i_2}{\cos r}R\ . \qquad (1)$$

Likewise, in $\triangle OC'D'$, the distance from O to entry C' satisfies

$$\frac{y_{c'}}{\sin i'_2} = \frac{R}{\sin(90^\circ - r)}\ ,\ \text{so}\quad y_{c'} = \frac{\sin i'_2}{\cos r}R\ . \qquad (2)$$

Varying i changes refraction r and surface incidence i_2, i'_2. At the critical angle given by

$$i_{20} = \arcsin(1/n) \approx 41.8^\circ \qquad (3)$$

total reflection occurs. Substitute $i_2 = i'_2 = i_{20}$ into (1), (2) to obtain

$$y_{oc} = y_{oc'} = \frac{\sin i_{20}}{\cos r}R\ , \qquad (4)$$

Thus

$$d = 2y_{oc} = 2\frac{\sin i_{20}}{\cos r}R\ . \qquad (5)$$

If $y_c > y_{oc}$ and $y_{c'} > y_{oc'}$, the rays enter, but reach the curved surface at angles above the critical angle i_{20}, so cannot exit. r increases with i. From (4), at $r \to 0$, i.e. $i \to 0$ (normal incidence), d is minimal:

$$d_{\min} = 2R\sin i_{20} = 1.33\,R\ . \qquad (6)$$

At $i \to 90^\circ$ (grazing), $r \to 41.8^\circ$. Substitute $r = 41.8^\circ$ into (4) to find $d_{\max} = 1.79\ \ R$. (7)

2. In Fig. 2, φ is the Oz - OD angle; φ' is the Oz - OD' angle. At total reflection,

$$\varphi = i_{20} + r\ , \qquad (8)$$

$$\varphi' = i_{20} - r\ , \qquad (9)$$

and

$$\theta = \varphi + \varphi' = 2i_{20} \approx 83.6^\circ\ . \qquad (10)$$

Thus θ is independent of i, i.e. θ of i. At grazing

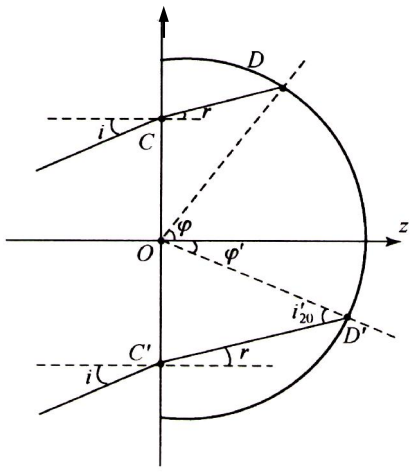


Fig. 2

incidence, $i \approx 90°$, $r = 41.8°$. From （8）, （9）,

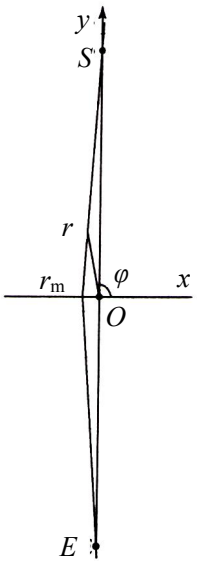
$$\varphi = 83.6° \ , \ \varphi' = 0°. \qquad (11)$$

6.

The equation

$$r = \frac{GM/c^2}{a\cos\varphi + a^2(1 + \sin^2\varphi)} \qquad (1)$$

is even in φ; the ray is symmetric about the polar axis. Left of the origin, $a < 0$; on the right, $a > 0$. The figure shows $a < 0$, with axis Ox and dwarf at O. In （1）, insert the periapsis coordinates $r = r_m$, $\varphi = \pi$ and note $a^2 \gg |a|$, giving



$$a \approx -GM/c^2 r_m \ . \qquad (2)$$

Rays past both sides of the dwarf subtend θ_S at the observer; this can be expressed in different ways, yielding different solutions.

Method 1: let d be the dwarf－Earth distance. Approximately,

$$\theta_S \approx 2r_m/d \ . \qquad (3)$$

In （1）, substitute observer coordinates $r = d$, $\varphi = -\pi/2$:

$$a^2 \approx GM/2c^2 d \ . \qquad (4)$$

Eliminate a from （2） and （4） to find

$$r_m = \sqrt{2GMd/c^2} \ . \qquad (5)$$

Substitute （5） into （3）:

$$\theta_S \approx \sqrt{8GM/c^2 d} \ ; \qquad (6)$$

Thus $$M \approx \theta_S^2 c^2 d/8G \ , \qquad (7)$$

Here $d = 3.187 \times 10^{17}$ m. Numerical substitution gives

$$M \approx 2.07 \times 10^{30} \text{kg} \ . \qquad (8)$$

Method 2: coordinates of the ray going to infinity can be written

$$r \to \infty \ , \ \varphi = -\frac{\pi}{2} + \frac{\theta}{2} \ . \qquad (9)$$

Take $\theta_S \approx \theta$, insert （9） into （1）,(set the （1） denominator to zero; note $\theta \ll 1$:

$$a\theta/2 + 2a^2 = 0 \ .$$

Thus $$\theta_S \approx \theta = -4a = \sqrt{8GM/c^2 d} \ , \qquad (10)$$

Use （4） and note $a < 0$. Equations （10）(and （6） agree, again giving （8）.

Method 3: angle θ_S equals the angle between the ray tangents at the observer:

$$\sin\frac{\theta_S}{2}=\frac{\triangle(r\cos\varphi)}{\triangle r}=\cos\varphi-r\sin\varphi\frac{\triangle\varphi}{\triangle r}\ . \qquad (11)$$

From the ray equation (1), calculate $\triangle\varphi/\triangle r$:

$$\sin\frac{\theta_S}{2}=\cos\varphi-r\sin\varphi\frac{GM/c^2}{r^2a\sin\varphi}=\cos\varphi-\frac{GM}{c^2ra}\ ;$$

Insert observer coordinates $r=d$, $\phi=-\pi/2$ and a from (4), noting small θ_S:

$$\theta_S\approx\frac{2GM}{c^2d}\sqrt{\frac{2c^2d}{GM}}=\sqrt{\frac{8GM}{c^2d}},$$

This agrees with (6), again giving (8).

Method 4: use (2) to rewrite (1) as

$$-r_m=r\cos\varphi-\frac{GM}{c^2r_mr}\left[(r\cos\varphi)^2+2(r\sin\varphi)^2\right],$$

Thus $$x=-r_m+\frac{GM}{c^2r_mr}(x^2+2y^2)\ . \qquad (12)$$

For $y\to-\infty$, the asymptote of (12) is

$$x=-r_m-\frac{2GM}{c^2r_m}y\ .$$

This straight line has x intercept $-r_m$ and slope

$$\frac{1}{-2GM/c^2r_m}\approx\frac{1}{-\tan(\theta_S/2)}\approx-\frac{1}{\theta_S/2}\ .$$

Thus $\theta_S\approx 4GM/c^2r_m$. Substitute (5) for r_m to obtain (6), and hence (8).

7.

1. (I) Helium has two electrons; E^+ is the energy to remove one to infinity, satisfying $He\quad E^+\to He^++e^-$. For E^+, find the ion energy after one electron is ionized, E^*: one electron orbiting the helium nucleus. Here are two methods.

Method 1: in the mechanical equation

$$\frac{2ke^2}{r^2}=\frac{mv^2}{r}$$

r is the radius and v the speed. Bohr's ground-state quantization ($\hbar$) gives

$$r_0=\hbar^2/2ke^2m\ . \qquad (1)$$

Thus the helium-ion energy is

$$E^*=\frac{p_0^2}{2m}-\frac{2ke^2}{r_0}=-\frac{2k^2e^4m}{\hbar^2}\ , \qquad (2)$$

Here p_0 is ground-state electron momentum. Substitution gives

$$E^* = -\frac{2(ke^2)^2mc^2}{(\hbar c)^2} \approx -54.4\ \text{eV} . \qquad (3)$$

Ignoring electron interaction, the helium ground energy E_0 is 2 times this value:

$$E_0 = 2E^* \approx -108.8\ \text{eV} . \qquad (4)$$

The ion energy E^* minus atomic ground energy E_0 is the first ionization energy:

$$E^+ = E^* - E_0 = -E^* \approx 54.4\ \text{eV} . \qquad (5)$$

Method 2: the helium-ion energy is

$$E^* = \frac{p^2}{2m} - \frac{2ke^2}{r} .$$

Insert the ground angular momentum $rp = \hbar$ to rewrite (3):

$$E^* = \frac{\hbar^2}{2mr^2} - \frac{2ke^2}{r} = \frac{\hbar^2}{2m}\left(\frac{1}{r} - \frac{2ke^2m}{\hbar^2}\right)^2 - \frac{2k^2e^4m}{\hbar^2} .$$

Minimum ground energy makes the first term of (4) zero. Thus radius and energy

$$r_0 = \frac{\hbar^2}{2ke^2m} , \quad E^* = -\frac{2k^2e^4m}{\hbar^2}$$

are respectively as in (1), (2).

(II) Here are two methods for helium ground energy E_0 and radius r_0.

Method 1: the mechanical equation

$$\frac{mv^2}{r} = \frac{2ke^2}{r^2} - \frac{ke^2}{(2r)^2} = \frac{7ke^2}{4r^2}$$

and ground-state quantization $rmv = \hbar$ give the radius

$$r_0 = 4\hbar^2 / 7ke^2m , \qquad (6)$$

Thus the helium ground energy is

$$E_0 = 2\left(\frac{p_0^2}{2m} - \frac{2ke^2}{r_0}\right) + \frac{ke^2}{2r_0} = -\frac{49k^2e^4m}{16\hbar^2} ; \qquad (7)$$

Numerical substitution gives

$$E_0 = -\frac{49(ke^2)^2mc^2}{16(\hbar c)^2} \approx -83.4\ \text{eV} , \qquad (8)$$

$$r_0 = \frac{4(\hbar c)^2}{7ke^2mc^2} \approx 0.0302\ \text{nm} .$$

Therefore the helium first ionization energy is

$$E^+ = E^* - E_0 \approx 29.0\ \text{eV} . \qquad (9)$$

This still exceeds the measured helium ionization energy 24.6 eV by 4.4 eV .

Method 2: the helium atom energy is

$$E = 2\left(\frac{p^2}{2m} - \frac{2ke^2}{r}\right) + \frac{ke^2}{2r} = \frac{\hbar^2}{mr^2} - \frac{7ke^2}{2r}$$

This can be rewritten as

$$E = \frac{\hbar^2}{m}\left(\frac{1}{r} - \frac{7ke^2m}{4\hbar^2}\right)^2 - \frac{49k^2e^4m}{16\hbar^2} .$$

The first term on the right vanishes at minimum energy. The ground energy and radius

$$E_0 = -\frac{49k^2e^4m}{16\hbar^2} , \quad r_0 = \frac{4\hbar^2}{7ke^2m}$$

are respectively as in （7），（ 6）.

2.（ I） The particle crosses the lead plate upward. Its charge is positive.

（II） Velocity v is normal to the field; Lorentz force is in the paper. Magnetism changes momentum direction, not magnitude. Neglecting gas drag, in the steady field the path is a circular arc of constant radius. The equation of motion is

$$qBv = \left|\frac{\triangle p}{\triangle t}\right| = \frac{p\triangle\phi}{\triangle t} = \frac{pv}{r}, \qquad (1)$$

Here q is charge, v speed and p momentum; $\triangle\phi$ is the angle turned in time $\triangle t$, and r is the curvature radius. Hence

$$p = qBr . \qquad (2)$$

$q = e$; let p_d, p_u be entering and exiting momenta. In （1），substitute the data to obtain

$$p_d = 63.0\ \text{MeV/c} , \quad p_u = 22.8\ \text{MeV/c} . \qquad (3)$$

For $pc \gg mc^2$, use special relativity. From

$$p = \frac{mv}{\sqrt{1-(v/c)^2}} . \qquad (4)$$

we obtain

$$v = \frac{c}{\sqrt{1+(mc/p)^2}} . \qquad (5)$$

Let v_d, v_u be entering and exiting speeds. Insert （2） and $m = 0.51\ \text{MeV}/c^2$ into （3） to obtain

$$v_d \approx c , \quad v_u \approx c . \qquad (6)$$

The mean speed through lead is $\bar{v} = (1/2)(v_d + v_u) \approx c$ $\triangle t$ is the transit

time. Then

$$\bar{v}\cos\theta\triangle t = d \ . \qquad (7)$$

Let $\triangle p_{du}$ be the momentum change magnitude. The mean force on the plate is

$$f = \frac{\triangle p_{du}}{\triangle t} = \frac{p_d - p_u}{d/(\bar{v}\cos\theta)} \approx \frac{(p_d - p_u)\, c\cos\theta}{d} \ ; \qquad (8)$$

Substitution gives

$$f \approx 1.04 \times 10^{-9}\ \mathrm{N} \ . \qquad (9)$$

（III） The time for one particle to cross the plate is

$$\triangle t = \frac{d}{\bar{v}\cos\theta} \approx \frac{d}{c\cos\theta} \approx 2.07 \times 10^{-11}\ \mathrm{s} = 0.0207\ \mathrm{ns} \ , \qquad (10)$$

far shorter than pulse period $\tau = 2\mathrm{ns}$. The mean force on the plate during the pulse passage is

$$F \approx (p_d - p_u) j \ ; \qquad (11)$$

Substitution gives

$$F = 0.107\ \mathrm{N} \ . \qquad (12)$$

Using （4）, the particle energy can be written

$$E = \sqrt{p^2c^2 + m^2c^4} \ , \qquad (13)$$

Thus the particle energies before and after crossing are

$$E_d = \sqrt{p_d^2c^2 + m^2c^4} = 63.0\ \mathrm{MeV} \ , \ E_u = \sqrt{p_u^2c^2 + m^2c^4} = 22.8\ \mathrm{MeV} \ . \qquad (14)$$

The heat absorbed by the plate during the pulse is therefore

$$Q = (E_d - E_u) j \tau \ ; \qquad (15)$$

Substitution gives

$$Q = 8.04 \times 10^{-2}\ \mathrm{J} \ . \qquad (16)$$

$$r = f\cot(2\alpha) \tag{2}$$

Combining the two equations gives

$$x = R\sin\left(\frac{1}{2}\arctan\frac{f}{r}\right)。 \tag{3}$$

Scoring: correct ray diagram, 5 points; equation (3) , points.